

Ankul Kumar

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📍 Bangalore

🐙 Github [↗](#)

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Experience

Machine learning Intern [↗](#)

Humrobo technology

Designed and developed a scalable movie recommendation system leveraging **Collaborative Filtering** and **Content-Based Filtering algorithms** to provide personalized user experiences.

Developed a **scalable backend** using Python and Django, implementing **RESTful APIs** for seamless data flow. Optimized **API response time by 20%** through **code optimization** and query performance tuning.

Created a responsive and dynamic front-end interface with **HTML, CSS, JavaScript**, and integrated it with the backend to deliver an engaging user experience.

Deployed the application on **Heroku**, leveraging **microservices architecture** and **Scalability Engineering techniques** to enhance **performance and fault tolerance**. Integrated **CI/CD pipelines**, reducing **deployment time by 50%** and ensuring **seamless updates without downtime**.

Conducted **rigorous unit testing and algorithm validation** to ensure recommendation accuracy, achieving a **15% increase in average session time** and improving overall user engagement.

Machine Learning Trainee (Internshala) [↗](#)

Designed and implemented a scalable web scraping architecture to efficiently extract and process large volumes of data from diverse online sources while adhering to **ethical web scraping best practices** and **data privacy regulations**.

Built a **scalable backend** using Python and Flask, focusing on **modular architecture** for seamless integration. Implemented **unit testing** to enhance system reliability and reduce downtime by **30%**.

Implemented advanced features, including **error handling, retry mechanisms**, and **dynamic data parsing**, to enhance reliability and accuracy during data collection.

Optimized **data collection processes**, reducing **data processing time by 40%** and improving **accuracy by 25%** using advanced **error handling, retry mechanisms**, and **dynamic data parsing**.

Profile

Python Developer | Backend & AI Engineer experienced in building scalable RESTful APIs using Flask and Django, with a strong foundation in FastAPI, semantic search (RAG), and AI/ML model deployment. Skilled in designing microservices architectures, optimizing inference pipelines, and integrating vector databases using LangChain and embeddings. Proficient in asynchronous Python, containerization (Docker), and CI/CD practices for high-performance, production-ready applications.

Skills

Languages: Python, JavaScript, SQL

Frameworks & Backend: FastAPI, Django, Flask, DRF

Frontend: HTML, CSS, JavaScript, AngularJS

AI/ML & RAG: Scikit-learn, TensorFlow, LangChain, Embeddings, Semantic Search, Vector Stores

DevOps & Cloud: Docker, Git, GitHub Actions, Heroku, AWS

Databases: PostgreSQL, MongoDB, MySQL, Redis

Concepts: RESTful APIs, Async Programming, Microservices, Model Serving, CI/CD, Unit Testing, Software Design Patterns

Tools: BeautifulSoup, Selenium, SQLAlchemy, Airflow (Basics), VS Code

Education

Bachelor of Technology in Computer Science And Artificial Intelligence

NIET, Greater Noida [↗](#)

07/2020 – 06/2024

Achievements

Successfully solved complex coding problems on GeeksforGeeks, showcasing strong analytical skills. [↗](#)

Demonstrated expertise in data structures and algorithms through challenges on HackerRank. [↗](#)

Certifications:

- Python Developer Associate (ISO 14001)
- Machine Learning Specialization (coursera)

Projects

Quizify

- Developed a **scalable quiz** platform using Django, Improved API response time by 30% while handling 5,000+ daily quiz submissions.
- **Integrated MYSQL Database:** Designed and optimized database models in MYSQL to ensure seamless data handling, storing user progress, quiz results, and question banks efficiently.
- **Responsive Frontend Development:** Implemented a responsive and engaging user interface using Django's built-in templating system, HTML, CSS, and JavaScript, providing a seamless experience for quiz navigation and real-time feedback.